

Research papers

A time series physics-informed neural network framework for the state of health estimation of lithium-ion batteries

Weina Zhang , Hao Zhang ^{*}, Zhongqin Bi

College of Computer Science and Technology, Shanghai University of Electric Power, Shanghai, 201306, China

ARTICLE INFO

Keywords:

Lithium-ion batteries

SOH estimation

Physics-informed neural network

ABSTRACT

The state of health (SOH) of lithium-ion batteries (LIBs) is a key parameter of the battery management system. The SOH estimation method based on physics-informed neural network (PINN) jointly models the battery degradation mechanism with deep-learning models, effectively solving problems such as the lack of interpretability existing in traditional data-driven methods. However, the existing SOH estimation methods based on PINN have not yet paid attention to the temporal dynamic characteristics of LIBs. This paper proposes the Time Series Physics-Informed Neural Network (TS-PINN) framework, which integrates the advantages of temporal characteristics, battery aging mechanism and data-driven models to achieve accurate estimation of SOH. This method adopts a tandem structure design of dual neural networks to jointly construct a unified model for battery capacity attenuation dynamics and deep-learning. The model captures the temporal characteristics of LIBs through Long Short-Term Memory (LSTM), enhancing its ability to capture the long-term dependencies of the aging process. For the complex degradation process of LIBs, a capacity degradation model is constructed using partial differential equations to guide the training of neural networks. Besides, this paper proposes an adaptive weight adjustment algorithm based on homoscedasticity uncertainty, achieving autonomous collaborative balance of the weights of each loss term. The experimental results on the TJU, the XJTU and the NASA datasets show that, compared with other models, this method demonstrates significant advantages in the estimation accuracy of SOH.

1. Introduction

With the advancement of global energy transition and sustainable development strategies, lithium-ion batteries (LIBs) have become an indispensable core of energy storage in modern society due to their outstanding advantages such as high energy density, long cycle life, low self-discharge rate and environmental friendliness [1–3]. From portable electronic devices to electric vehicles (EVs), and then to large-scale grid energy storage systems (ESS), the application breadth and depth of LIBs are expanding at an unprecedented speed [4–6]. However, the performance of LIBs will undergo irreversible decline over charge and discharge cycles and time. This process is known as battery aging [7]. The external manifestations of aging mainly include the attenuation of maximum available capacity and the increase of internal impedance, which affect the device's endurance and power output performance. The aging process directly influences its safety, reliability and economy.

As a key technology to ensure the safe and efficient operation of batteries, one of the core tasks of the battery management system (BMS) is to precisely monitor and predict the state of health (SOH) of the battery [8–10]. SOH is a key indicator used to quantify the degree of

battery aging. Accurate SOH estimation not only provides users with reliable information on remaining battery range or standby time, but also guides BMS to adopt the optimal charge and discharge strategy to delay battery degradation [11]. More importantly, it can warn of potential safety risks, such as thermal runaway [12], which is crucial for avoiding catastrophic accidents [13]. Therefore, accurately estimating SOH in BMS is of vital importance, which is of great significance for enhancing the full life cycle value of batteries and ensuring system safety.

1.1. Literature review

In previous studies, many methods and strategies for SOH estimation have been mentioned and summarized. Liu et al. [14] summarized the SOH estimation method based on electrochemical impedance spectroscopy (EIS), discussed the influence of frequency-domain feature extraction, temperature and state of charge (SOC) on EIS, as well as the advantages and limitations of EIS in online/offline scenarios. Ren et al. [15] provided a comprehensive methodological review for

^{*} Corresponding author.

E-mail address: leon_hao@mail.shiep.edu.cn (H. Zhang).

SOH estimation, comparing the properties and applicable scenarios of feature engineering, model-driven, and data-driven methods, and pointing out future research directions. In addition, Li et al. [16] conducted a systematic review of LIBs SOH and remaining useful life (RUL) prediction methods based on artificial neural networks, covering various paradigms such as shallow networks, deep networks, and hybrid structures. Generally speaking, they can be roughly divided into three categories: direct measurement methods, model-based methods, and data-driven methods.

The direct measurement method mainly calculates SOH directly by measuring the specific physical and chemical parameters of the battery [17]. Among them, electrochemical impedance spectroscopy and ampere-hour integration are two common direct measurement methods. Electrochemical impedance spectroscopy (EIS) infers the aging state by applying a small sinusoidal alternating current signal and measuring its voltage response [14]. The ampere-hour integration method directly calculates capacity by precisely measuring the current and time during the charging and discharging process. These methods usually require expensive and bulky dedicated equipment, and the measurement process is time-consuming, making it difficult to integrate into BMS that need real-time response. Therefore, they are not suitable for online application scenarios.

Model-based methods mainly estimate SOH by establishing physical or chemical models that can describe the aging behavior of batteries. These models mainly include the Equivalent Circuit Model (ECM) and the Electrochemical Model (EM). ECM generally uses components such as resistors and capacitors to simulate the external dynamic characteristics of batteries. They have a simple structure, low computational load and are easy to implement in BMS. Chen et al. [18] proposed a new method that combines ECM with deep learning networks. The results show that, compared with other alternative methods, the proposed scheme can achieve higher prediction accuracy. Thomas et al. [19] proposed a method for predicting the capacity and health status of lithium-ion batteries based on the Simplified Equivalent Circuit Model (ER-ECM). This method significantly reduces the computational load required for modeling, with an error reduction of 40% compared to the method using all model parameters. Hou et al. [20] proposed a linear time-varying parameter-varying (LTPV) model in response to the limitations of traditional ECM in capturing the internal reactions and external characteristics of batteries. This method demonstrates stronger online adaptability and aging modeling capabilities. The pseudo-two-dimensional model (P2D) and single-particle model (SPM) in EM, based on the porous electrode theory and concentration polarization theory, can more accurately describe the lithium-ion migration, diffusion and reaction kinetics processes inside the battery [21,22]. Researchers have developed various filtering algorithms to couple these models for precise SOH estimation. Although EM has a high fidelity, its computing cost is extremely high, making its online application difficult to achieve. Overall, model-based methods still need to overcome several practical challenges to meet the reliability requirements of BMS in actual operation.

In recent years, with the development of artificial intelligence and big data technologies, data-driven methods represented by machine learning and deep learning have made remarkable progress in the field of SOH estimation [23]. The data-driven approach regards the battery as a “black box” [24], directly learning the mapping relationship between SOH and measurement data such as voltage, current, and temperature from a vast amount of historical monitoring data, without the need to deeply understand the complex internal mechanism of the battery. For instance, Yao et al. [25] proposed a deep learning framework that uses generative adversarial network (Gan) for data augmentation and feature mapping to enhance the accuracy of SOH estimation for lithium-ion batteries. The experimental results show that the accuracy of the samples is improved after data augmentation, and the average absolute error is less than 2%. Zhao et al. [26] designed

a novel Transformer-Embedded model that combines multi-scale attention mechanisms with temporal convolutional networks (TCN) to achieve joint estimation of SOH and SOC. Mamba is an innovative state space model architecture specifically designed for handling sequential data and has also gained considerable popularity in the field of SOH research. Wang et al. [27] proposed a multi-forward step prediction method for LIBs, which integrates segmented gated recurrent unit (SGRU) and DualMamba, achieving high-precision predictions on NASA and MIT datasets while also featuring low computational complexity. In addition, Wang et al. [28] constructed a framework that integrates wavelet-FFT frequency-domain decomposition and sparse attention enhanced Mamba to address the challenges of multivariate time series analysis of LIBs, achieving significant improvements in the prediction accuracy, stability, and generalization of SOH. Although these methods can effectively handle the highly nonlinear problems during the battery degradation process. However, pure data-driven methods also have their inherent limitations. It is precisely because of this “black box” characteristic that pure data-driven models may violate physical principles during the prediction process, resulting in predicted values that are neither realistic nor reasonable, and making the model lack physical interpretability.

To address the issue of the lack of interpretability in pure data-driven models, the emerging method of physics-informed neural network (PINN) has emerged. The basic principle of this method is to combine physical principles with data-driven approaches, and guide and correct the training and learning process of neural networks through prior knowledge in the field of physics. In the field of LIBs research, researchers are also exploring the application of this method. Zhu et al. [29] proposed a Bayesian calibrated PINN framework for SOH estimation of retired LIBs, and the superiority of this method was verified on a self-built dataset. Zhang et al. [30] constructed a physics-informed dual neural network (PIDNN) by integrating physics-based electrochemical models with deep learning models to estimate electrochemical parameters and lithium ion concentrations in the solid phase and electrolyte, thereby calculating battery capacity attenuation. Wei et al. [31] proposed a novel PINN that integrates the attention mechanism with the Kolmogorov–Arnold network (KAN) to capture the dynamic features of battery degradation and enhance SOH prediction. Li et al. [32] proposed a method for estimating the state of charge (SOC) of LIBs that integrates physical information mechanisms and Mamba, named physics-informed mamba neural network (PIMNN), which effectively improves the accuracy and interpretability of SOC estimation. Seo et al. [33] proposed the Butler-Volmer informed neural network (BVINN), which directly embeds the classical Butler-Volmer equation into the network loss function. This method can maintain higher performance even in an environment with scarce data and improve the interpretability of SOH estimation. Although all the above research methods integrate the prior knowledge of LIBs degradation, their application not only requires researchers to have a deep understanding of the degradation principle of LIBs, but also none of them have studied the temporal characteristics of the battery data itself. The above-mentioned problems not only restrict the use of PINN by researchers outside the field of electrochemistry, but also to some extent restrict its applicability and flexibility.

1.2. Contributions

To make up for the deficiencies of the above-mentioned research and optimize the limitations under various conditions. This paper proposes the TS-PINN framework for the accurate estimation of LIBs SOH. Unlike the current PINN method, this method explores the ability of classic health characteristics and LIBs time series property characteristics in the field of LIBs capacity degradation research to estimate SOH, and its performance has been tested on battery datasets under different chemical materials and temperature conditions. Specifically, the method proposed in this paper utilizes the incremental capacity (IC)

Table 1

Detailed information of each batch of batteries.

Dataset	Batch	Chemical component	Nominal capacity (mAh)	Cut-off voltage (V)	Temperature (°C)	Number of cells
TJU	1	$\text{Li}_{0.86}\text{Ni}_{0.86}\text{Co}_{0.11}\text{Al}_{0.03}\text{O}_2$	3500	2.65–4.2	25,35,45	66
	2	$\text{Li}_{0.84}\text{Ni}_{0.83}\text{Co}_{0.11}\text{Mn}_{0.07}\text{O}_2$	3500	2.5–4.2	25,35,45	55
	3	Blend of 42 (3) wt.% LiNiCoMnO_2 and 58 (3) wt.% LiNiCoAlO_2	2500	2.5–4.2	25	9
XJTU	1–6	$\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$	2000	2.5–4.2	Room temperature	55
NASA	B0005	$\text{LiNi}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$	2000	2.7–4.2	24	1
	B0006	$\text{LiNi}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$	2000	2.5–4.2	24	1
	B0007	$\text{LiNi}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$	2000	2.2–4.2	24	1
	B0018	$\text{LiNi}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$	2000	2.5–4.2	24	1

features of the constant current charging stage and extracts the features with temporal characteristics from the original experimental data with the help of the Long Short-Term Memory (LSTM) module. These mixed features are jointly input into the constructed network model. During the model training stage, this method models battery degradation, uses a neural network to fit the capacity degradation of the battery, and embeds it as a physical constraint into the loss function of the model to guide the training and learning of the neural network. The specific contributions of this article are as follows:

(1) To enhance the model's ability to handle nonlinear, dynamic and long-term dependency issues during the LIBs aging process, on the basis of extracting features using IC curves, the LSTM module is introduced to extract the time series features from the original battery data, enabling it to effectively learn and remember the degradation history information of the battery during long cycle periods. Improve the modeling accuracy of the model for the dynamic process of battery aging.

(2) Based on the mathematical principle of homoscedastic uncertainty, this paper proposes an adaptive weight adjustment algorithm named AdpAdj. This algorithm can dynamically and automatically balance the contributions of each loss term during the training process, solve the problem of manual parameter adjustment, and improve the training efficiency and convergence stability of the model.

(3) To achieve a deep integration of data-driven and physical mechanism, this paper constructs a series structure of dual neural networks. The former is used to output the preliminary SOH estimate value, while the latter is used to characterize the capacity degradation model, improving the physical interpretability and prediction accuracy of the model. Experiments conducted on the TJU dataset, the XJTU dataset and the NASA dataset verified that this method has superior SOH estimation performance compared to other network models.

1.3. Organization of the paper

The remaining chapters of this article are arranged as follows. Section 2 introduces the research methods. Section 3 provides a detailed elaboration of the experiments and analyses. Section 4 summarizes the full text.

2. Research methodology

The TS-PINN framework for LIBs SOH estimation is proposed in this study to enhance the accuracy and interpretability of SOH estimation., as shown in Fig. 1. Firstly, this study conducts a correlation analysis between the battery charge and discharge cycle periods and the IC curve. Subsequently, IC features that can reveal the battery aging process are extracted from the IC curve, and the time series features in the battery cycle experiment data are extracted through LSTM. Finally, the constructed TS-PINN model framework adopts a tandem structure design of dual neural networks, which are respectively used to estimate SOH and simulate battery capacity degradation and convert them into physical constraint embedding loss functions to guide the neural network training process. In addition, the training process of the model adopts a multi-objective loss function, which includes three key

loss terms. The time cost of manually adjusting parameters is extremely high. This study designs an adaptive weight adjustment algorithm to optimize the training process.

2.1. Feature extraction from IC curves

To verify the applicability of the proposed method, this study synthesized aging experimental data from 189 batteries of different types and chemical compositions, mainly including the TJU battery dataset publicly disclosed by Zhu et al. [34] The XJTU battery dataset provided by Wang et al. [35] And the NASA battery dataset [36]. The battery information of the three datasets is shown in Table 1.

The TJU dataset contains three types of batteries :66 NCA batteries with a nominal capacity of 3500 mAh and a cut-off voltage of 2.65–4.2 V, 55 NCM batteries with a nominal capacity of 3500 mAh and a cut-off voltage of 2.5–4.2 V, and 9 batteries with a nominal capacity of 2500 mAh. NCM+NCA batteries with a cut-off voltage of 2.5–4.2 V (represented by batch1, batch2 and batch3 respectively). These batteries were subjected to cyclic tests in test environments with different temperatures (25 °C, 35 °C and 45 °C) and different charging currents (0.25 C to 4 C).

The XJTU battery dataset uses 55 18650 model batteries produced by Lishen Company. Their main materials are $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$, with a nominal capacity of 2000 mAh and a nominal voltage of 3.6 V. The charge and discharge cut-off voltages of the batteries are 4.2 V and 2.5 V respectively. This dataset uses six charge–discharge schemes, including fixed charge–discharge, random discharge with fixed current at different cycles, random walk, and the charge–discharge strategy for geosynchronous orbit satellites (using batch1 to batch6 to represent these six charge–discharge schemes respectively), and conducts cyclic experiments at room temperature until battery failure.

The NASA battery dataset is obtained from the NASA Ames Predictive Center of Excellence (PCoE). The experimental batteries are 18,650 batteries produced by A123 systems company, with a rated capacity of 2 Ah, and the experimental temperature is 24 °C. In this paper, B0005 battery, B0006 battery, B0007 battery and B0018 battery are selected for research. In the charging phase, each battery is charged with 1.5 A constant current (CC) to 4.2 V, and then charged with 4.2 V constant current (CV) to less than 20 mA. In the discharge phase, each cell is discharged at 2 A CC until the voltage reached 2.7 V, 2.5 V, 2.2 V and 2.5 V, respectively.

The IC curve contains information about the battery aging mechanism and can provide a deep understanding of the battery's aging process. Dubarry et al. [37] conducted cycle life tests on commercial rechargeable LIBs and analyzed the causes of battery capacity attenuation using the ICA method. Gismero et al. [38] developed and verified a new method for measuring battery SOH points, and simultaneously studied the ability of the ICA method to estimate the SOH of real electric vehicles. The overall root mean square error of the estimated battery SOH was 2%. They found that the peaks, peak areas and other features of the IC curve would evolve in a certain pattern as the battery aged. Ansean et al. [39] found through experiments that the loss of active substances in LIBs would cause a reduction in battery capacity, resulting in a year-on-year decrease in the end-of-discharge

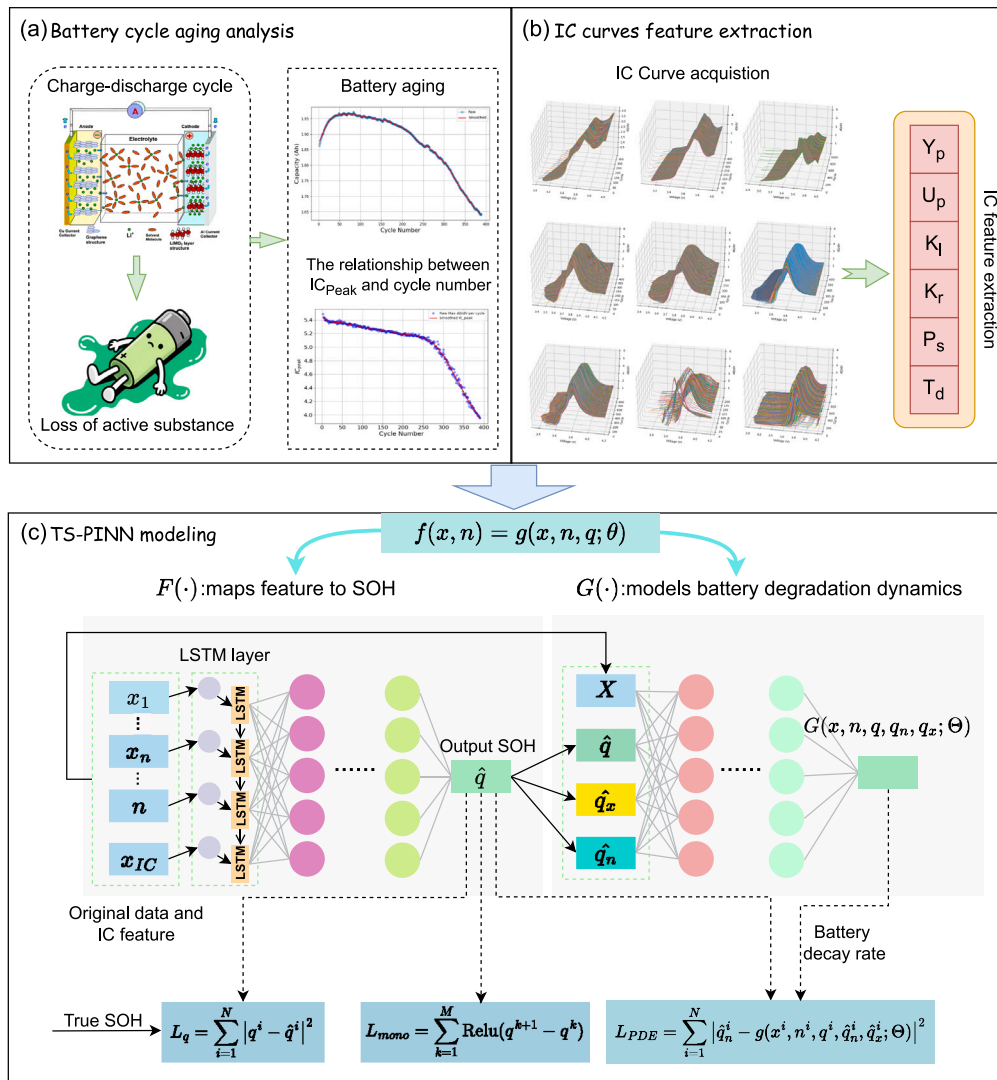


Fig. 1. The proposed TS-PINN framework for estimating SOH of lithium-ion batteries. (a) With the increase of the number of charge and discharge cycles, the active substances in the LIBs are gradually depleted, resulting in a decrease in capacity. There is a linear relationship between the peak IC and the number of cycles, from which the health index is extracted. (b) For feature extraction, we use IC curve data for each cycle, including IC peak (Y_p), peak position (U_p), left and right slopes (K_l , K_r), peak area (P_s), and firing time (T_d). The left panel shows the 3D IC curves for the battery datasets. (c) In the proposed TS-PINN, both IC features and raw battery data are input. LSTM extracts the time series features in the original data, where q and $\hat{q}$ represent the true value and the estimated SOH, n and x represent the period and feature, i is the sample index, and n and x subscript are partial derivatives. The functions $f(\cdot)$ and $g(\cdot)$ map the characteristics to the SOH and describe the battery degradation dynamics.

voltage, ultimately leading to a steeper slope on the right side of the IC curve. Meanwhile, the formation of the solid electrolyte interface (SEI) would hinder the reaction between lithium ions and the anode, accelerating the voltage increase during charging. This leads to the left slope of the IC curve becoming “gentle” [40]. Therefore, this paper conducts relevant research under the condition that a complete IC curve containing enough information can be obtained, and features are extracted from it as part of the model input to perform SOH estimation.

The IC curve is a widely used analytical tool in the field of battery capacity degradation research, which is obtained by differential calculation of the voltage and capacity data collected during the constant current charging process of batteries [41]. It describes the incremental change in capacity over a continuous voltage range during the constant current charging stage of the battery. The mathematical calculation formula of the IC curve can be expressed as:

$$Q = It \quad (1)$$

$$IC = \frac{dQ}{dU} = I \frac{dt}{dU} \quad (2)$$

where Q denotes the battery capacity, I represents the charging current, t is the charging time, and U stands for the battery voltage.

Generally, during the cyclic experiments on LIBs, there will be a large amount of measurement noise, which will directly affect the research and analysis of batteries. As shown in Fig. 2(a), the initial IC curve that has not been smoothed has significant high-frequency noise, which seriously affects the extraction and quantitative analysis of IC features. The original data has a large number of random fluctuations within the voltage range of 3.4 V–4.2 V, making it impossible to identify any valid information. Therefore, it is necessary to smooth the curve. To reduce the influence of noise interference, this study uses a finite difference with a fixed voltage width to interpolate the initial IC curve. The effect is shown in Fig. 2(b). The specific mathematical formula is as follows:

$$\frac{dQ}{dU} \approx \frac{\Delta Q}{\Delta U} = \frac{Q_t - Q_{t-1}}{U_t - U_{t-1}} \quad (3)$$

where Q_t and Q_{t-1} represent the battery charge capacity at time t and $t - 1$ respectively, and U_t and U_{t-1} denote the battery charging voltage

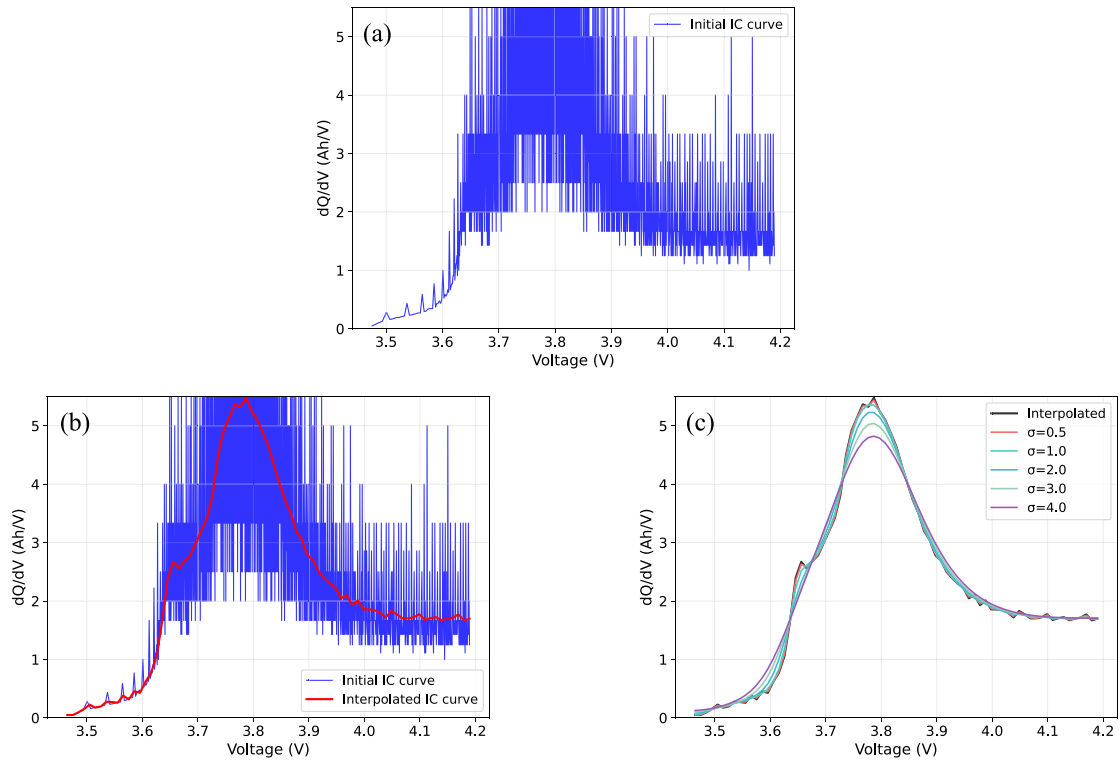


Fig. 2. The initial IC curve and the IC curves smoothed by the finite difference method and Gaussian filtering with different parameters.

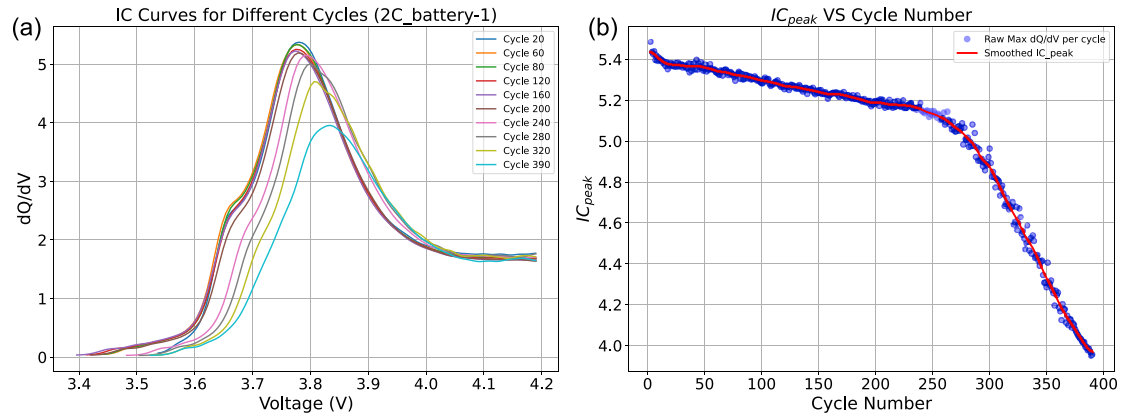


Fig. 3. Changes in IC curve and the relationship between IC_{Peak} and cycle number: (a) Changes of IC curve with cycle number. (b) The relationship between IC_{Peak} and cycle number.

at time t and $t - 1$ respectively. It can be clearly seen from the figure that the IC curve after interpolation and smoothing effectively reduces the discontinuity of the data. The interpolated IC curve is smoother than the original IC curve, filtering out noise while retaining important information to the greatest extent. However, there is still a certain amount of noise in the current curve.

To further reduce the interference of noise on the IC curve, Gaussian filtering (GS) is adopted to smooth the curve. GS filters are commonly used in smoothing curves and can separate low-frequency signals from high-frequency noise. The GS filter has the characteristic of Gaussian distribution and can be expressed by the following equation:

$$G(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \quad (4)$$

where μ is the average value and σ is the standard deviation of the control window size. When using the GS filter, each data point can be replaced by the weighted average of its adjacent data points. Therefore,

the most recent data has a greater impact on the average value, while the more distant data plays a less significant role. When using Gaussian for smoothing, μ is usually set to 0 because we want each piece of data to be the one that has the greatest impact on its new value after smoothing. The smoothing effect is influenced by the σ value. The larger the σ value, the smoother the curve. However, if σ is too large, the curve may lose some important information. Therefore, the value of σ should be within an appropriate range. Fig. 2(c) shows the influence of different σ values on the smoothing effect of the IC curve. When $\sigma=0.5$, the curve still retains a lot of detailed information, but the noise suppression effect is limited. When the σ value is 2.0, 3.0 and 4.0, the smoothness of the curve gradually increases, but the key features of the curve will change. When $\sigma=1.0$, the curve effectively suppresses noise while maintaining detailed features. Therefore, in this study, a GS filter with an σ value of 1 was chosen to smooth the curve.

IC curves provide assistance in studying the mechanism of battery performance degradation by analyzing the evolution of electrochemical

processes within batteries. Battery aging is typically characterized by a gradual decline in battery performance, mainly due to the combined effect of complex chemical reactions such as the degradation of electrode materials, the thickening of the SEI film, and the loss of electrolyte. These internal aging mechanisms are clearly characterized on the IC curve, specifically manifested as a decrease in peak height, a shift in peak position, an increase in peak width, etc. By analyzing the changes in the above-mentioned curve characteristics, the degree of battery aging can be effectively identified. Therefore, IC curve analysis is regarded as a key technology for studying the aging mechanism of LIBs, providing an important basis for battery health management.

To gain a more intuitive understanding of the changing trend of battery aging on the IC curve, this study took the batteries in the first batch of the XJTU dataset as an example and analyzed their IC curves during the constant current charging stage, as shown in Fig. 3, with a fixed voltage width ΔU of 0.01 V adopted [42]. As can be seen from Fig. 3(a), the peak value of the IC curve will gradually decrease and shift to the right as the number of battery charge and discharge cycles increases, and the curve trend will also become increasingly gentle. The decrease in the peak value of the curve is closely related to the chemical reactions occurring inside the battery. As the number of charge and discharge cycles increases, the chemical reactions inside gradually intensify, and the active substances are gradually degraded, thereby leading to a decline in battery capacity. Therefore, there is a monotonic relationship between the peak value of the IC curve and the number of charge and discharge cycles, as shown in Fig. 3(b). Based on this, the features extracted from the IC curve in this study include: IC peak (y_p), peak position (U_p), left peak slope (k_l), right peak slope (k_r), peak area (P_s), and discharge time (T_d). The horizontal and vertical coordinates of the peak of the IC curve are U_p and y_p respectively. The three characteristics of k_l , k_r and T_d can be calculated by the following formula:

$$U_l = U_p - \frac{\Delta U}{2}, \quad U_r = U_p + \frac{\Delta U}{2} \quad (5)$$

$$k_l = \frac{y_p - y_l}{U_p - U_l}, \quad k_r = \frac{y_p - y_r}{U_p - U_r} \quad (6)$$

$$T_d = T_r - T_l \quad (7)$$

where T_r represents the discharge time at voltage U_r , and T_l denotes the discharge time at voltage U_l .

2.2. The proposed model description

2.2.1. Long short-term memory network module

RNNs are often used to handle time series data, but when dealing with long-term dependent sequences during training, they may encounter problems such as vanishing gradients and exploding gradients [43]. LSTM can effectively solve these problems, and its specific structure is shown in Fig. 4. The LIBs cyclic experimental data have strong temporal characteristics and essentially fall within the category of time series data. Based on the advantages of LSTM in dealing with long-term series dependency problems, this study introduces LSTM to enhance the model's modeling ability for time series dependency and pay more attention to the historical information of battery degradation. This method introduces the LSTM module in the model feature input stage. Besides inputting the extracted IC features, it also inputs the original experimental data of the battery into the LSTM layer to obtain the time series features. Subsequently, the two types of features will jointly act on the SOH estimation.

Specifically, LSTM dynamically regulates the storage and forgetting of information through gating mechanisms, namely input gates, forget gates, and output gates, and extracts dynamic information related to battery degradation. To fully explore the temporal dependence in the battery degradation process, the sliding window method is used to conduct segmented sampling of the original battery data and input it

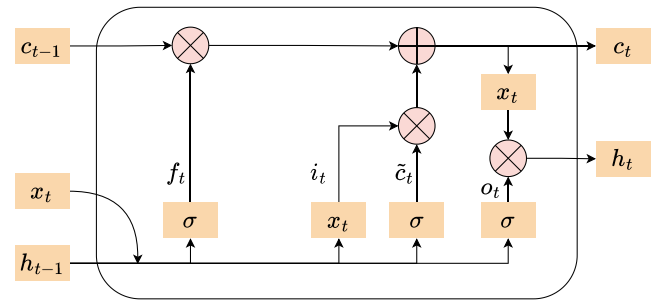


Fig. 4. The core structure of LSTM.

into the LSTM layer for extracting temporal features. The length of the sliding window is set to 10 time steps (For the selection of window length, please refer to Section 3.5), each of which contains the original information in the dataset. The step size is set to 1 to maximize data utilization. Therefore, the samples received by LSTM each time include all the cyclic data within 10 consecutive experimental time points in the dataset, in order to precisely capture the long-term and short-term dependencies during the battery degradation process. The specific internal calculation process of the LSTM unit is as follows.

Firstly, the forget gate will calculate, based on the current input x_t , which includes information such as current, voltage, capacity, temperature, power, and time in the dataset, as well as the previous hidden state h_{t-1} , which important information should be retained and which should be “forgotten”. The specific formula is as follows:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (8)$$

Next, the input gate also determines how much of the current input battery cycle experiment data x_t and the previous hidden state h_{t-1} will be written to the memory unit. At the same time, a candidate value $\tilde{c}_t$ for a memory unit will also be calculated, which represents the potential contribution of new information at time step t . The calculation formulas for both are as follows:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (9)$$

$$\tilde{c}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (10)$$

LSTM will update the state of the memory unit by combining the output of the forgetting gate, the output of the input gate, and the candidate values of the memory unit. The specific formula is:

$$C_t = f_t * C_{t-1} + i_t * \tilde{c}_t \quad (11)$$

Finally, the output gate will calculate the current hidden state h_t based on the original battery data x and the previous hidden state h_{t-1} . The calculation process is as follows:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (12)$$

$$h_t = o_t * \tanh(C_t) \quad (13)$$

In LSTM, the short-term memory chain h_t and the long-term memory chain C_t are two key components. Their interaction and update enable LSTM to achieve the storage and processing of long-term series data. Here, W represents different weight matrices, b represents different computational biases, and σ and $\tanh$ respectively represent the Sigmoid and Tanh activation functions.

For each input window, the model selects the hidden state of the last time step of the LSTM output sequence as the timing feature of the window. This feature vector integrates the dynamic information of all time steps within the window and can effectively represent the degradation state of the battery during this time period. The temporal features will be concatenated with the IC features and input into the subsequent linear layers of the model to complete the SOH estimation task.

2.2.2. Modeling battery degradation

Battery aging usually manifests externally as the number of charge and discharge cycles increases, such as capacity degradation and increased internal resistance. To describe the battery aging mechanism more accurately, researchers have constructed various empirical models from different perspectives to express the functional relationship between battery capacity loss and usage time or cycle period, including linear models [44], exponential models [45,46], etc. The models proposed by predecessors merely described the battery capacity degradation trajectory as a single-variable function relationship related to time, without fully reflecting the complex relationships behind battery aging. In fact, battery aging is not only related to time, but also closely associated with factors such as operating temperature, charge and discharge rates, and depth of discharge. Ref. [47] established a multivariate functional relationship between the battery aging rate and current, voltage, temperature, and reactant concentration, and constructed a coupled aging model. Compared with the empirical model mentioned earlier, it can more accurately describe the actual degradation process of the battery. Therefore, modeling the battery capacity degradation trajectory as a simple time function is insufficient. In this study, it is modeled as a multivariate functional relationship:

$$q = f(x, n) \quad (14)$$

here, x represents the vector composed of the extracted IC features and the timing features output by the LSTM module, and n represents the number of battery charge and discharge cycles.

Under the condition of not losing generality, to better describe the kinetics of battery degradation, its capacity attenuation rate can be expressed as:

$$\frac{\partial q}{\partial n} = g(x, n, q; \theta) \quad (15)$$

The above formula is an explicit PDE parameterized by θ , where $g(\cdot)$ is a nonlinear function with the independent variable x , n , and q . The function $g(\cdot)$ is used to describe the kinetics of battery degradation. By altering the representation of this nonlinear function, it can be expressed as the degradation of various factors. The linear model, exponential model, and power-law model mentioned can all be regarded as special cases of Eq. (15).

In fact, the exact mathematical expression of $g(\cdot)$ is unknown and difficult to calculate. To address the above challenges, Wen et al. [48] proposed a semi-physical and semi-empirical dynamic model based on the Verhulst model to capture the attenuation trend of battery SOH. When there is little prior knowledge of dynamics, they utilized the deep Hidden physical model (DeePHPM) to discover potential control dynamic models. Chen et al. [49] modeled battery capacity degradation by combining the Verhulst model with the Arrhenius equation to obtain the exact expression. The paper models the attributes that affect battery capacity attenuation from the perspective of empirical degradation models and state space equations, and uses neural networks to capture the dynamics of battery degradation. It is proposed to approximate this nonlinear function $g(\cdot)$ using a generalized function approximator $g'(\cdot)$ with a parameter of θ_0 . Thus, Eq. (15) converts into:

$$q_n \approx g'(x, n, q, q_n, q_x; \theta_0) \quad (16)$$

where $q_n = \frac{\partial q}{\partial n}$, this method uses a neural network $F(x, n; \Phi)$ and its learning parameters Φ to $f(x, n)$ model, and use the automatic differential mechanism to calculate q_n . $q_x = \left[\frac{\partial q}{\partial x_1}, \frac{\partial q}{\partial x_2}, \dots, \frac{\partial q}{\partial x_n} \right]^T$ represents the first-order partial derivative of the capacity q with respect to the feature x . Considering the overall computational complexity and training complexity, this method now only considers the influence of the first-order partial derivative on it, while ignoring the higher-order derivatives above the second order. The proposed scheme can effectively combine the fitting ability of neural networks and the physical interpretability of mechanism models, and its advantages are very significant. For the function approximator $g'(\cdot)$, Used this method with learning parameters

Θ neural network $G(\cdot)$ modeling so that it can from x , n and q_x learn the aging mechanism of the battery.

Based on the above research analysis, a capacity degradation study of PINN for LIBs was constructed:

$$\Lambda := \frac{\partial F(x, n; \Phi)}{\partial n} - G(x, n, q, q_n, q_x; \Theta) \quad (17)$$

where $\frac{\partial F(x, n; \Phi)}{\partial n}$ represents the partial derivatives of the neural network $F(\cdot)$ used to output SOH with respect to n , and $G(x, n, q, q_n, q_x; \Theta)$ represents the dynamic equation of battery capacity degradation modeled by the neural network.

The calculation formula for battery SOH can be defined as:

$$q^k = f(x, k) = \frac{Q^k}{Q^0} \quad (18)$$

where Q^k represents the battery capacity with a cycle count of k , and Q^0 is the rated capacity of the battery. When $k = n$, the SOH value of the battery should satisfy the fitting result of the neural network $F(\cdot)$. At sample i , $\Lambda(x^i, n^i) = 0$ should be used to approximate Eq. (15). Therefore, in the training process of neural networks, the specified PDE loss must be met:

$$L_{\text{PDE}} = \sum_{i=1}^N |\Lambda(x^i, n^i)|^2 \quad (19)$$

where N represents the total number of samples, and i represents the data of the i th sample. The loss term L_{PDE} quantifies the extent to which the predicted values of the neural network satisfy the laws of physics. By minimizing this loss, the network is compelled not only to fit the data but also to conform to the potential physical laws of battery degradation. This, as a powerful regularization mechanism, significantly enhances the generalization ability of the model. This constraint ensures that the learned SOH decay trajectory is controlled by a consistent physical model, thereby enhancing the reliability and interpretability of the results. In addition, the loss function also includes the true data loss, as shown in Eq. (20), and the monotonicity loss, as shown in Eq. (21).

$$L_q = \sum_{i=1}^N |q^i - \hat{q}^i|^2 \quad (20)$$

$$L_{\text{mono}} = \sum_{k=1}^M \text{ReLU}(\hat{q}^{k+1} - \hat{q}^k) \quad (21)$$

here, $\hat{q}$ represents the estimated SOH value, M represents the total number of cycles, and ReLU is Rectified Linear Unit function. The monotonicity constraint that L_{mono} is SOH should conform to the monotonically decreasing nature of the battery degradation process (unless capacity increase occurs), and ReLU acts as a penalty function. By minimizing this loss and enforcing soft constraints, the model is guided to learn the monotonic non-increasing function of SOH throughout the cyclic lifecycle. It not only ensures the physical rationality of the prediction but also makes the model more robust to measurement noise.

The formula for the total loss function is as follows:

$$L = \lambda_q L_q + \lambda_{\text{PDE}} L_{\text{PDE}} + \lambda_{\text{mono}} L_{\text{mono}} \quad (22)$$

Among it, λ_q , λ_{PDE} and λ_{mono} are the weight parameters of each loss term.

2.3. Weight coefficients tuning in training process

Generally, in neural network training, the predicted values learned by the network training are extremely sensitive to the proportion of weights of each loss term [50], and the distribution strategy of the loss function weights directly affects the model performance. This is even more so in this study. Under normal circumstances, the weights of each loss term in the loss function are manually adjusted, which heavily

relies on expert experience. This approach has many problems, such as being time-consuming and labor-intensive, and when the model is learning different datasets, it needs to manually readjust the weights to achieve the best effect, lacking generalization. Wang et al. [51] discovered that the numerical stiffness during the PINN training process would lead to an unbalanced backpropagation gradient during the model training period, and proposed an algorithm that adjusts the weight coefficients accordingly based on the gradient statistics. In addition, Bischof et al. [52] proposed a novel PINN adaptive loss balancing scheme based on random backtracking, named ReLoBRaLo.

This method proposes an adaptive weight adjustment algorithm based on homoscedasticity uncertainty, named the AdpAdj algorithm. This algorithm converts the weight coefficients into learnable parameters through probabilistic modeling, achieving dynamic adjustment of the weights of each loss term and alleviating the problem of gradient imbalance, thereby addressing the numerous drawbacks brought about by manual parameter adjustment. While ensuring the predictive performance of the model, Shorten the training duration and improve the training efficiency.

In this study, the estimated SOH value $\hat{q}$ output by the neural network has homoscedasticity uncertainty with the true SOH value q . It can be assumed that the posterior probability of the true SOH value q follows a Gaussian distribution with σ^2 as the variance:

$$p(q | F(x, n; \Phi)) = \mathcal{N}(F(x, n; \Phi), \sigma^2) \quad (23)$$

Then the negative logarithmic likelihood of Eq. (23) is:

$$-\log p(q | F(x, n; \Phi)) \propto \left(\frac{1}{2\sigma^2} \|q - \hat{q}\|^2 + \log \sigma^2 \right) \quad (24)$$

here, σ represents the noise of the real data. When the true value q of the battery SOH and the estimated value $\hat{q}$ of SOH are known, σ can be adaptively learned during the training process, and its setting is related to the weight coefficients of each loss term in Eq. (22). Based on Eq. (22) and (24), the multi-objective negative log-likelihood function can be calculated:

$$\begin{aligned} & -\log p(q, PDE, \text{mono} | F(x, n; \Phi), G(x, n, q, q_n, q_x; \Theta)) \\ & = \lambda_q L_q + \lambda_{PDE} L_{PDE} + \lambda_{\text{mono}} L_{\text{mono}} + \log \lambda_q \lambda_{PDE} \lambda_{\text{mono}} \end{aligned} \quad (25)$$

To ensure the stability of the values during the training process and prevent problems such as division by zero or gradient explosion during backpropagation, we define $\lambda = -\log \lambda$ and substitute it into Eq. (25) to obtain a new loss function:

$$\begin{aligned} L = & \exp(-\lambda_q) L_q + \exp(-\lambda_{PDE}) L_{PDE} \\ & + \exp(-\lambda_{\text{mono}}) L_{\text{mono}} + \lambda_q + \lambda_{PDE} + \lambda_{\text{mono}} \end{aligned} \quad (26)$$

where λ is a learnable parameter, the TS-PINN is trained by minimizing Eq. (26) to balance the multiple losses during training.

3. Experiments and analysis

The TS-PINN model constructed in this paper uses six extracted IC features, raw experimental data as input to estimate battery SOH. In order to eliminate the influence of feature scale differences on model performance, this study uses max-min normalization method to scale all features to the numerical interval of $[-1, 1]$. For the TJU and XJTU datasets, the training set, validation set and test set are divided in a ratio of 6:2:2, and the test samples are evenly distributed on the experimental datasets to ensure the fairness of the experiment. For the NASA dataset, the model is trained using leave-one-out cross-validation, where each battery can be used as a training and test set, Table 2 shows the detailed model parameter Settings. To comprehensively evaluate the performance of the proposed TS-PINN model, three comparison models are used: Multi-layer Perceptron (MLP) [53] with the same network structure and number of parameters, convolutional

Table 2

The parameter settings in TS-PINN.

Hyperparameters	Value
Number of layers	6
Number of neurons	128
Batch size	512
Number of epochs	200
Dropout probability	0.2
Initial learning rate	0.01

neural Network (CNN) [53], and the PINN model constructed by Wang et al. [35] (hereafter referred to as baseline).

The model is trained with an Adam optimizer with an initial learning rate of 0.01. If the validation loss does not improve over 20 consecutive epochs, the learning rate drops by 0.5 times. Set the batch size to 512 and the maximum number of training cycles to 200. To prevent overfitting of the model, a dropout mechanism was introduced during the training process, with the dropout rate set to 0.2. The model was trained 200 times, and 10 training and testing experiments were conducted in each batch of the dataset to demonstrate its stability. All the experiments in this section were conducted on devices with an Intel Core i5-10300H CPU, 16 GB RAM and NVIDIA GTX 1650Ti GPU.

3.1. The evaluation criteria

To evaluate the effectiveness of the proposed framework, this paper conducts a comprehensive assessment of the SOH estimation error using the mean absolute percentage error (MAPE) and the root mean square error (RMSE). The specific calculation formulas are as follows.

$$\text{MAPE} = \frac{1}{N} \sum_{i=1}^N \frac{|\hat{q}_i - q_i|}{q_i} \times 100\% \quad (27)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{q}_i - q_i)^2} \quad (28)$$

where $\hat{q}_i$ represents the estimated SOH value of the battery in the i th cycle, q_i represents the true SOH value of the battery in the i th cycle, and N represents the total number of cycles.

3.2. Comparison with other models

3.2.1. The experiments on TJU and XJTU datasets

The SOH estimation performance of the TS-PINN model proposed in this paper in three batches of the TJU dataset and six batches of the XJTU dataset is shown in Fig. 5. The model can maintain an excellent level on both datasets. In the TJU dataset, the model has the best estimation effect on batch3, and most of the scatter points are distributed near the diagonal, and the mean absolute error can be maintained at 0.0051. For batch1 and batch2, although the model is not as excellent as batch3, it can also maintain relatively good performance, with the mean absolute error of 0.0117 and 0.0114. In the XJTU dataset, the mean absolute error of the model can be maintained in a relatively good range. Most of the scatter points of the six batches are distributed near the diagonal, but there are also some scattered points far away. The best and worst estimation performance of batch3 and batch6 are 0.0047 and 0.0084.

The detailed performance of the TS-PINN model and other comparison models on the two datasets is shown in Table 3. Overall, the TS-PINN model demonstrates significant superiority in all evaluation indicators. The physical constraint terms in the model guide the neural network to learn the inherent degradation laws of the battery rather than merely fitting the surface features of the data, which greatly reduces the risk of the model generating predictions that violate physical intuition. Compared with pure data-driven MLP and CNN models, TS-PINN has achieved an order of magnitude accuracy advantage in

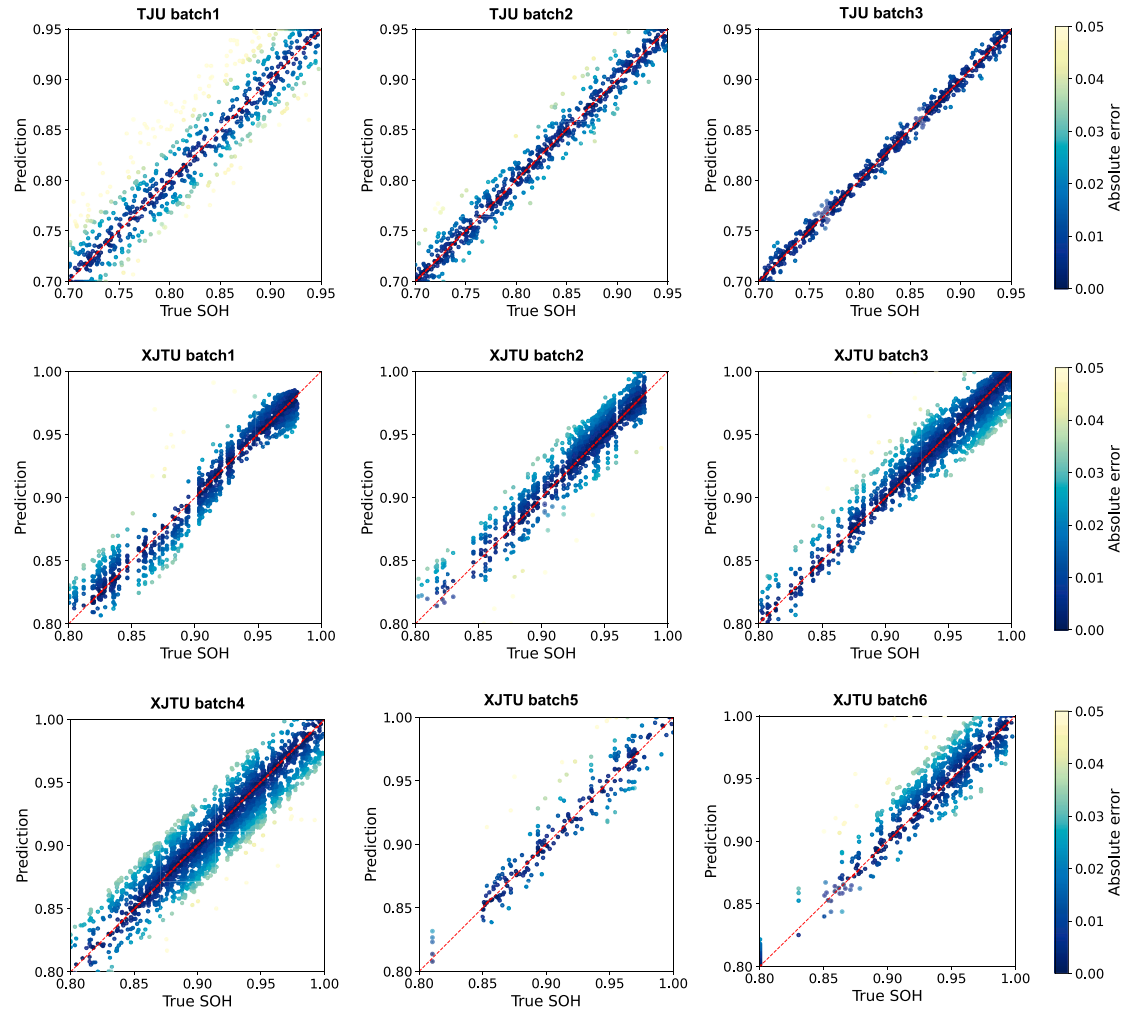


Fig. 5. The fitting effect of the proposed TS-PINN for SOH estimation on two datasets.

Table 3

The experimental results of TS-PINN and other three comparison models on TJU and XJTU datasets. The bold is the best value.

Dataset	Batch	TS-PINN			PINN [35]			MLP [53]			CNN [53]		
		MAPE	RMSE	Time (s)	MAPE	RMSE	Time (s)	MAPE	RMSE	Time (s)	MAPE	RMSE	Time (s)
TJU	1	0.0121	0.0150	1354	0.0164	0.0158	1186	0.0174	0.0166	1287	0.0210	0.0220	1402
	2	0.0122	0.0138	1568	0.0119	0.0132	1424	0.0142	0.0149	1634	0.0142	0.0149	1586
	3	0.0035	0.0052	226	0.0080	0.0079	247	0.0116	0.0112	315	0.0130	0.0131	268
XJTU	1	0.0070	0.0081	201	0.0070	0.0094	136	0.0259	0.0287	231	0.0333	0.0401	217
	2	0.0057	0.0073	182	0.0113	0.0122	190	0.0296	0.0342	218	0.0277	0.0330	233
	3	0.0049	0.0063	194	0.0086	0.0100	173	0.0185	0.0217	262	0.0232	0.0276	169
	4	0.0069	0.0102	215	0.0071	0.0105	186	0.0203	0.0233	237	0.0189	0.0232	219
	5	0.0091	0.0110	168	0.0105	0.0135	131	0.0211	0.0261	204	0.0377	0.0492	185
	6	0.0110	0.0117	432	0.0063	0.0097	382	0.0179	0.0216	387	0.0151	0.0196	452

most batches, which strongly demonstrates the key role of physical information constraints in enhancing the generalization ability of the model.

Furthermore, the introduction of the LSTM module is another key factor for the success of the TS-PINN model. LIBs aging is a process with long-term dependence, and the current health status is influenced by the entire historical cycle. Compared with the baseline model, TS-PINN can effectively capture this dependency on the time series through the LSTM layer. The modeling ability of LSTM for historical information enables the model to better understand and predict the degradation trajectory under different strategies, thereby achieving stronger robustness.

It should be pointed out that due to the more complex model structure, TS-PINN has an increased running time compared to other models. However, this additional computational overhead is acceptable. By introducing the AdpAdj adaptive weight adjustment algorithm, the model can automatically allocate the weights of each loss term in different datasets and batches, avoiding the cumbersome manual parameter adjustment process. Compared with the fixed-weight strategy, this mechanism enables the model to maintain stable performance under different experimental conditions, significantly enhancing the efficiency and scalability of practical applications in the long run.

In the TJU dataset, the comprehensive advantages of the TS-PINN model are particularly prominent. Compared with the baseline model,

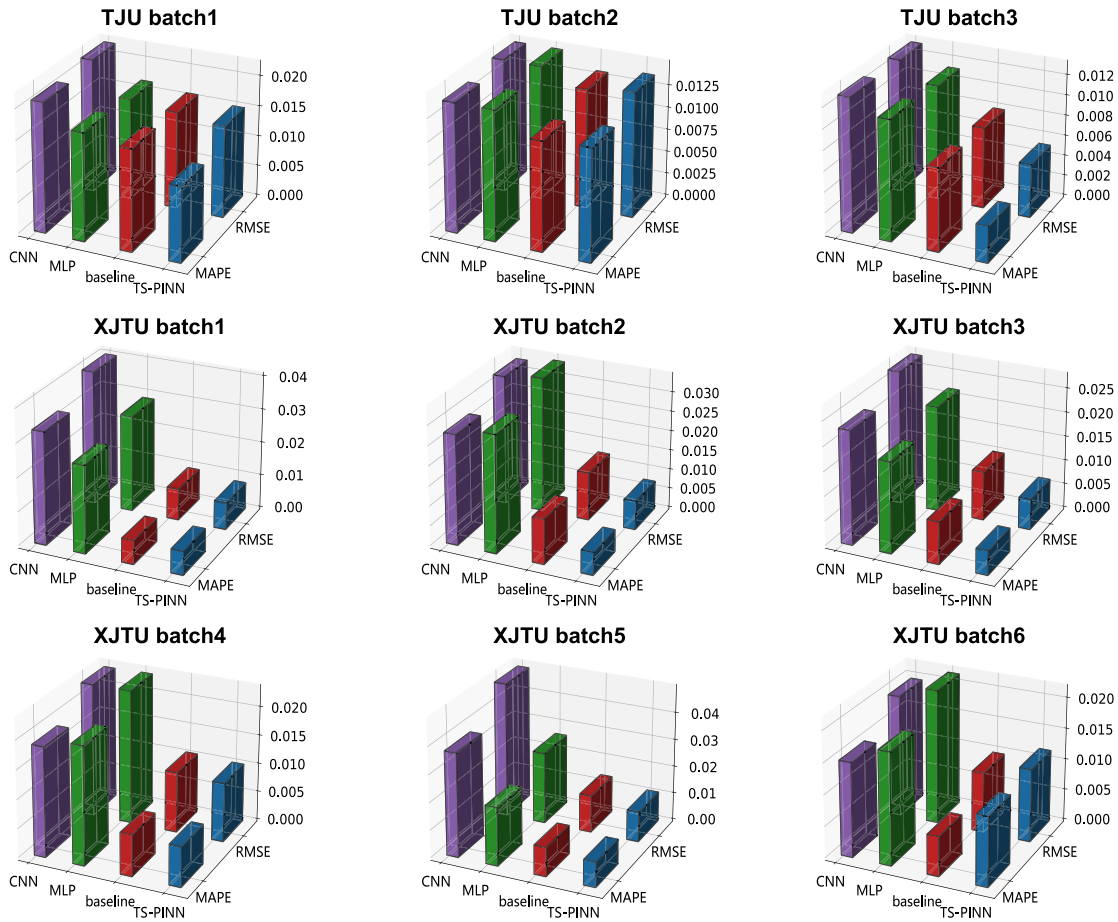


Fig. 6. Performance comparisons of different batches and different models.

its average MAPE and RMSE indicators have been optimized by 23.1% and 8.1% respectively. Especially in the third batch of tests, the MAPE and RMSE of the model were as low as 0.0035 and 0.0052 respectively, significantly outperforming all the comparison models. This strongly demonstrated its high efficiency and outstanding predictive ability under this dataset. Similarly, on the more complex XJTU dataset, the TS-PINN model also demonstrates strong robustness. Its average MAPE and RMSE have increased by 12.9% and 16.5% respectively compared to the benchmark model. For instance, in the third group of experiments, the MAPE and RMSE of the model were 0.0049 and 0.0063 respectively, which were far lower than those of other deep learning models. This verified the effectiveness of the strategy that combines time series analysis with physical models.

It is worth noting that there are exceptional cases in some batches of experiments. In the experiments of TJU batch 2 and XJTU batch 6, the performance of the baseline model was superior to that of TS-PINN. According to the preliminary analysis, this may be related to the NCM battery material characteristics used in TJU batch 2 and the charging–discharging strategy used in XJTU batch 6 to simulate the Geosynchronous Orbit (GEO) satellite. The appearance of these special cases does not deny the overall effectiveness of the TS-PINN model proposed in this paper, but clarifies the applicable boundary of the model and points out the specific direction for the next stage of research work, that is, how to further enhance the adaptability of the model to special materials and extreme working conditions.

The bar comparison of the performance evaluation metrics of the TS-PINN model with baseline, MLP and CNN, as well as the comparison of SOH degradation curves, are shown in Fig. 6 and Fig. 7, respectively. It can be intuitively seen that in each batch of the TJU dataset, the MAPE and RMSE bars of TS-PINN are the best or closest to the best,

and its SOH degradation curve is also the most consistent or nearly consistent with the real SOH degradation curve. Its estimation error is smaller, which indicates that the generalization ability of TS-PINN is stronger than that of the three comparison models. In different batches of the XJTU dataset, it can also be observed that in most cases, the RMSE and MAPE bars of TS-PINN are superior to those of the comparison models, and the SOH degradation curves are better than the prediction curves of other models. The above situations fully demonstrate that TS-PINN can effectively reduce estimation errors when dealing with data from different sources and batches, and its accuracy is superior to that of baseline and traditional MLP and CNN, reflecting the effectiveness and superiority of the TS-PINN model proposed in this paper in related tasks.

3.2.2. The experiments on NASA dataset

In order to verify the generalization of the proposed TS-PINN model on the SOH estimation task, we conduct an exhaustive comparison experiment on the NASA dataset with more capacity regeneration phenomena. Comparison of SOH estimation curves and error bars of TS-PINN and contrast models on four cells is shown in Fig. 8. The predicted curve of TS-PINN has the highest fit to the true SOH curve, which can accurately track the overall degradation trend and local capacity fluctuation of the battery. In contrast, the prediction curves of MLP and CNN show large deviations and significant noise interference. The superiority of TS-PINN is also further confirmed by the error histogram, which has the lowest MAPE and RMSE error bars in all datasets.

The detailed performance of TS-PINN model and other comparison models on NASA dataset is shown in Table 4. The experimental results clearly show that on all four test cells, the proposed TS-PINN model achieves the lowest values in both MAPE and RMSE, which

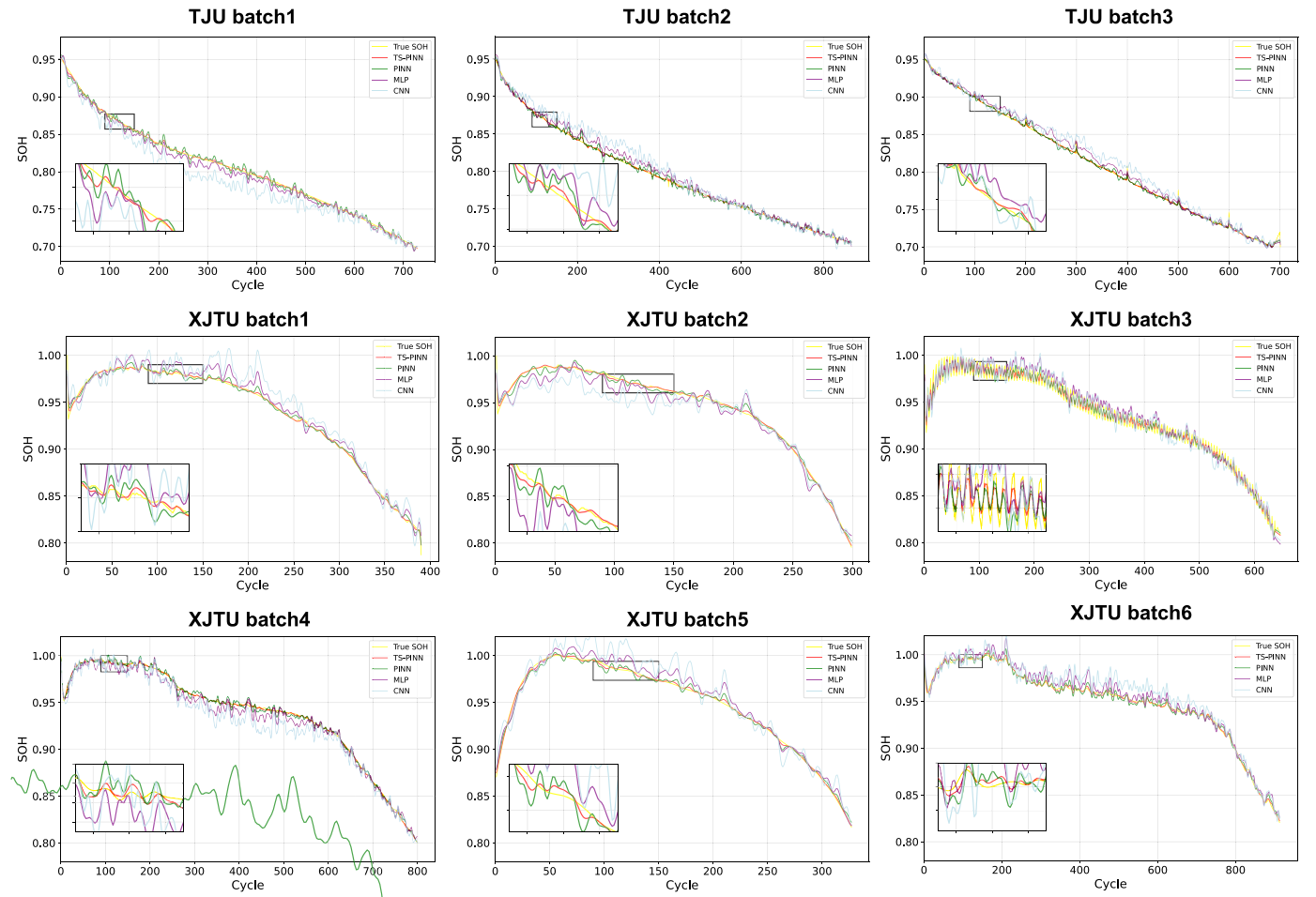


Fig. 7. Comparison of SOH degradation curves of different models on TJU and XJTU datasets.

Table 4

The experimental results of TS-PINN and other three comparison models on NASA dataset. The bold is the best value.

Dataset	Batch	TS-PINN			PINN [35]			MLP [53]			CNN [53]		
		MAPE	RMSE	Time (s)	MAPE	RMSE	Time (s)	MAPE	RMSE	Time (s)	MAPE	RMSE	Time (s)
NASA	B0005	0.0143	0.0138	83	0.0172	0.0156	73	0.0196	0.0188	91	0.0236	0.0231	98
	B0006	0.0161	0.0174	91	0.0210	0.0193	87	0.0323	0.0307	84	0.0319	0.0364	86
	B0007	0.0136	0.0130	78	0.0184	0.0162	76	0.0202	0.0300	87	0.0340	0.0391	71
	B0018	0.0153	0.0147	62	0.0176	0.0169	63	0.0228	0.0194	77	0.0380	0.0439	75

is significantly better than all comparison models. Taking the B0005 battery as an example, the MAPE and RMSE of TS-PINN are 0.0133 and 0.0128, respectively, which are better than baseline and other comparison models. Not only that, the TS-PINN model significantly improves the accuracy, but its computational cost is competitive with other models. For example, on the B0018 battery, TS-PINN has a running time of 62 s, which is the fastest among all models. On the B0005 battery, it takes 83 s less than MLP and CNN, but slightly more than baseline. On the B0007 battery, the computation time of all models is relatively close, and TS-PINN is at the same level as baseline. The experimental results fully demonstrate the high accuracy, high robustness, and reasonable computational efficiency of the proposed TS-PINN model in SOH estimation.

The TS-PINN model proposed in this paper demonstrates significant performance advantages and superior generalization ability over traditional data-driven models in the vast majority of test scenarios, especially when dealing with battery batches that exhibit complex nonlinear behaviors.

3.3. Analysis of adaptive weight adjustment mechanism

This study proposes an adaptive weight balance strategy to dynamically adjust the weight coefficients of each item in the multi-objective loss function. This mechanism quantifies the uncertainty of different loss terms by introducing a learnable uncertainty parameter σ , thereby achieving adaptive adjustment of weights. Due to space limitations, this paper selects the first four experiments of TJU batch1 and the first four experiments of XJTU batch1 for analysis. Fig. 9 shows the relative changes of the weight coefficients.

In the early stage of training, the data loss weight q , the partial differential loss weight pde , and the monotonicity loss weight $mono$ were all initialized to equal 1/3. This uniform initialization strategy ensures that the model does not lean towards any specific loss term in the early stage of training, providing a good starting point for subsequent adaptive adjustments. It can be observed from the image that q continuously rises throughout the training process, gradually increasing from the initial 1/3 to 0.4–0.5 at the end of the training, becoming the dominant weight. This phenomenon reflects that the

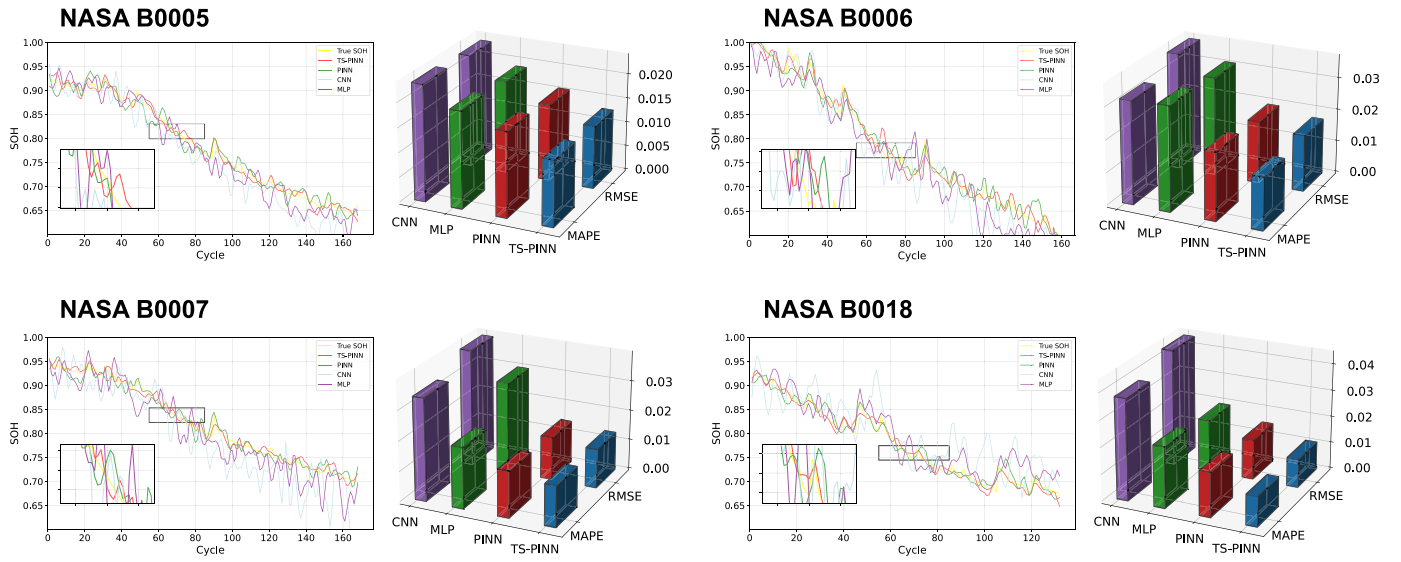


Fig. 8. The comparison of SOH curves and error bars of different models on NASA dataset.

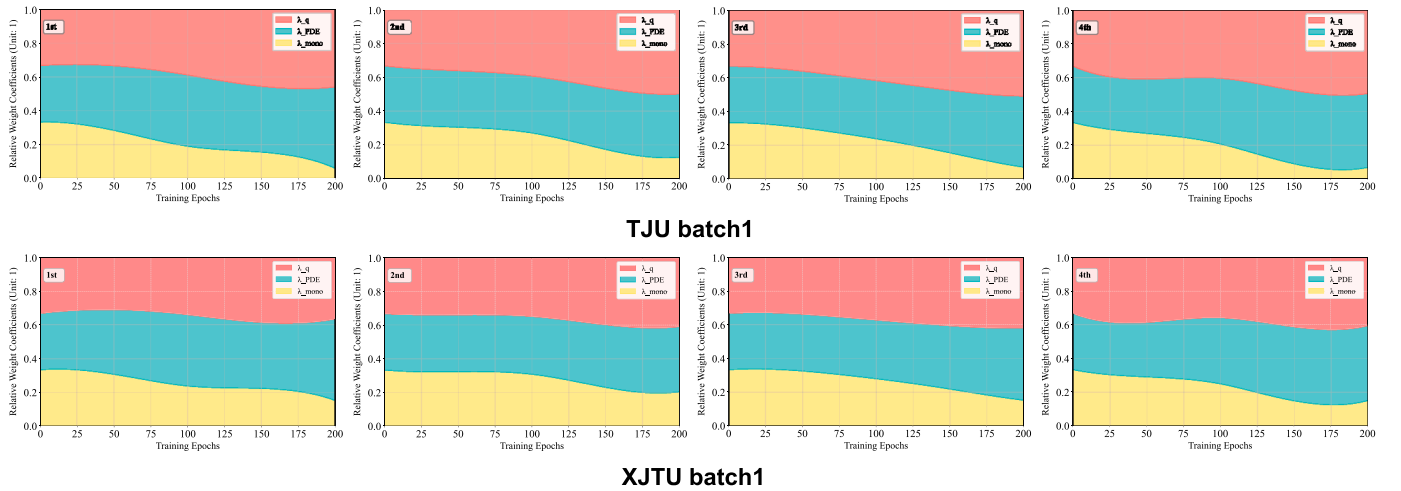


Fig. 9. The changes in the relative values of weight coefficients.

model gradually recognizes the importance of data fitting during the learning process. As the network parameters gradually converge to the optimal solution, the weight of the data loss term in the overall optimization objective naturally increases. pde showed a slow upward trend, rising from the initial 1/3 to around 0.45 in the end, maintaining a considerable contribution, indicating that physical constraints have always played an important role throughout the training process. The most significant change is reflected in mono, whose weight coefficient drops sharply during the training process, from the initial 1/3 to around 0.1 in the end. This sharp decline reflects the reduced relative importance of monotonicity constraints in the later stage of training. This might be because as the model gradually learns the physical law of battery capacity attenuation, the guiding role of monotonicity constraints gradually weakens. Meanwhile, the synergistic effect of the other two losses has been able to ensure the physical rationality of the prediction results relatively well, making the explicit monotonicity constraint relatively redundant.

As the training process progresses, the weight coefficients of the two batches exhibit significant and similar variation patterns. This stability demonstrates the robustness of the adaptive weight adjustment mechanism. The weight variation trajectories among different batches

and experiments are similar, indicating that this mechanism can stably identify and learn the relative importance of different loss terms without relying on specific initialization or randomness.

3.4. Ablation experiments

The previous text has verified the superiority of the TS-PINN model proposed in this paper over the three comparison models in estimation performance. This section introduces the ablation experiments of the LSTM module and the AdpAdj algorithm to prove the effectiveness and necessity of introducing these components. Table 5 presents in detail the evaluation index results of the ablation experiment. “without LSTM” represents the experimental results of PINN without an LSTM layer, and “without AdpAdj” represents the experimental results of TS-PINN without using the AdpAdj algorithm but with fixed weights.

The role of the LSTM layer can be evaluated by comparing the performance of the TS-PINN model with that of the “without LSTM” model. The experimental results show that after removing the LSTM layer, the prediction errors of the model on both datasets have increased to varying degrees. Taking batch2 of the XJTU dataset as an example, the MAPE and RMSE of the model lacking LSTM are

Table 5

The comparisons of ablation experiment results. The bold is the best value.

Dataset	Batch	TS-PINN		PINN [35]		without LSTM		without AdpAdj	
		MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE
TJU	1	0.0121	0.0150	0.0164	0.0158	0.0128	0.0152	0.0123	0.0152
	2	0.0122	0.0138	0.0119	0.0132	0.0125	0.0139	0.0123	0.0140
	3	0.0035	0.0052	0.0080	0.0079	0.0045	0.0078	0.0048	0.0071
XJTU	1	0.0070	0.0081	0.0070	0.0094	0.0072	0.0090	0.0071	0.0085
	2	0.0057	0.0073	0.0113	0.0122	0.0087	0.0116	0.0064	0.0081
	3	0.0049	0.0063	0.0086	0.0100	0.0059	0.0078	0.0063	0.0075
	4	0.0069	0.0102	0.0071	0.0105	0.0071	0.0103	0.0070	0.0105
	5	0.0091	0.0110	0.0105	0.0135	0.0102	0.0126	0.0096	0.0116
	6	0.0110	0.0117	0.0063	0.0097	0.0115	0.0130	0.0113	0.0121
NASA	B0005	0.0143	0.0138	0.0172	0.0156	0.0158	0.0161	0.0146	0.0152
	B0006	0.0161	0.0174	0.0210	0.0193	0.0179	0.0182	0.0193	0.0186
	B0007	0.0136	0.0130	0.0184	0.0162	0.0166	0.0159	0.0177	0.0165
	B0018	0.0153	0.0147	0.0176	0.0169	0.0163	0.0160	0.0168	0.0159

Table 6

The comparisons of the effects of different sliding window lengths. The bold is the best value.

Dataset	Batch	Length = 5			Length = 10			Length = 15			Length = 20		
		MAPE	RMSE	Time	MAPE	RMSE	Time	MAPE	RMSE	Time	MAPE	RMSE	Time
TJU	1	0.0159	0.0171	1288	0.0121	0.0150	1354	0.0137	0.0161	1433	0.0168	0.0183	1482
	2	0.0203	0.0177	1501	0.0122	0.0138	1568	0.0128	0.0152	1642	0.0215	0.0203	1687
	3	0.0091	0.0114	183	0.0035	0.0080	226	0.0069	0.0102	303	0.0121	0.0147	364
XJTU	1	0.0125	0.0101	156	0.0070	0.0081	201	0.0074	0.0093	287	0.0133	0.0115	315
	2	0.0098	0.0106	147	0.0057	0.0073	182	0.0071	0.0089	226	0.0123	0.0138	277
	3	0.0103	0.0124	142	0.0049	0.0063	194	0.0058	0.0061	265	0.0107	0.0110	327
	4	0.0087	0.0132	173	0.0069	0.0102	215	0.0056	0.0112	264	0.0098	0.0125	302
	5	0.0123	0.0173	128	0.0091	0.0110	168	0.0108	0.0126	197	0.0132	0.0153	235
	6	0.0160	0.0122	391	0.0110	0.0117	432	0.0116	0.0108	484	0.0172	0.0126	521

0.0087 and 0.116 respectively, which are significantly higher than those of the complete PINN model at 0.0057 and 0.0073. This strongly demonstrates that the LSTM network can effectively capture the long-term dependence and temporal dynamic characteristics in the capacity degradation of lithium batteries, which is crucial for improving the long-term prediction accuracy of SOH estimation. If this structure is lacking, the model will have difficulty learning the degradation law in the time dimension, thereby leading to a decline in prediction accuracy.

The experimental data shows that the MAPE and RMSE of the complete TS-PINN model using the AdpAdj algorithm in all test batches are lower than those of the model using fixed weights. For instance, on the batch5 of the XJTU dataset, the MAPE of the “without AdpAdj” model is 0.0096, while that of the TS-PINN model is 0.0091. Compared with fixed weights that require manual tuning, the adaptive adjustment strategy can guide the model to find a better gradient descent direction during the training process, avoiding one loss term dominating the training process while ignoring another, thereby enabling the model to converge to the optimal solution with stronger generalization ability.

3.5. Effect of sliding window length on model performance

To evaluate the impact of sliding window length on model performance, this study conducted comparative experiments with different window lengths on the TJU and XJTU datasets respectively. Four different sliding window lengths of 5, 10, 15 and 20 were set up in the experiment. MAPE and RMSE were used as evaluation indicators, and the training time was recorded simultaneously to analyze the computational efficiency. The detailed experimental results are shown in Table 6

It can be seen from the table that when the window length is 10, the model achieves the optimal MAPE and RMSE metrics in most batches, indicating that this length can effectively balance the ability to capture historical information and the efficiency of model training. In the batch3 of the TJU dataset, when the window length is 10, MAPE and RMSE are 0.0035 and 0.0080 respectively, which are significantly

better than other length Settings. Similarly, in the XJTU dataset, a window length of 10 also achieved the lowest error in most batches. As the window length increases, the training time shows an increasing trend. A window length of 10 ensures relatively high computational efficiency while maintaining performance, making it an ideal compromise choice.

The experimental results show that a sliding window length of 10 exhibits the best performance in most experimental scenarios and can effectively balance the capture of timing information and noise suppression. A sequence length that is too short cannot fully capture the temporal dependency during the battery aging process, while a sequence length that is too long introduces redundant information and noise, affecting the learning effect of the model. Based on the above analysis, this study selects a sliding window length of 10 as the standard configuration of the TS-PINN model.

4. Conclusions

In advanced BMS, long-term and high-precision estimation of LIBs SOH is crucial for ensuring system security, extending battery life and enhancing user experience. This paper proposes and verifies a TS-PINN framework for accurate estimation of SOH. In the feature extraction stage, the model not only utilizes the health features in the IC curve that can deeply reflect the electrochemical changes inside the battery, but also introduces the LSTM module to effectively extract the time series features from the original experimental data, enhancing the model's ability to handle nonlinear and long-term dependency issues during the battery aging process. In the model construction stage, this paper adopts a dual-neural network series structure design, which is respectively used for SOH estimation and fitting the approximate battery capacity degradation model, and embeds it as a physical constraint into the loss function to ensure that the model's prediction follows the basic physical laws of battery capacity degradation. In addition, in order to efficiently balance data loss, physical constraint loss and monotonicity loss, this paper designs an adaptive weight adjustment algorithm named AdpAdj. This algorithm is based on the principle of

homoscedasticity uncertainty and overcomes the cumbersome and inefficient traditional manual parameter adjustment. Through experiments conducted on the TJU dataset, the XJTU dataset and the NASA dataset, which include various battery types and experimental conditions, the superiority of the TS-PINN model proposed in this paper has been fully verified. Overall, this model outperforms traditional data-driven models and baseline models in the two key performance indicators of RMSE and MAPE. It demonstrated higher estimation accuracy and stronger generalization ability.

Although the TS-PINN model has demonstrated significant advantages in overall performance, its performance on a very few specific batches has not reached the optimal level. This phenomenon reveals the key directions and challenges for future research. Future research efforts will focus on delving into specific battery materials and special charging and discharging strategies to further enhance the robustness and generalization ability of the model. Meanwhile, the migration of the TS-PINN framework to other state estimation studies of LIBs, such as the estimation of RUL, is also within the scope of future research work.

CRedit authorship contribution statement

Weina Zhang: Writing – review & editing, Supervision, Conceptualization. **Hao Zhang:** Writing – original draft, Visualization, Software, Methodology. **Zhongqin Bi:** Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] F. Wu, J. Maier, Y. Yu, Guidelines and trends for next-generation rechargeable lithium and lithium-ion batteries, *Chem. Soc. Rev.* 49 (5) (2020) 1569–1614.
- [2] N. Nasajpour-Esfahani, H. Garmestani, M. Bagheritabar, D.J. Jasim, D. Toghraie, S. Dadkhah, H. Firoozeh, Comprehensive review of lithium-ion battery materials and development challenges, *Renew. Sustain. Energy Rev.* 203 (2024) 114783.
- [3] Y. Yang, R. Wang, Z. Shen, Q. Yu, R. Xiong, W. Shen, Towards a safer lithium-ion batteries: A critical review on cause, characteristics, warning and disposal strategy for thermal runaway, *Adv. Appl. Energy* 11 (2023) 100146.
- [4] J. Xu, X. Cai, S. Cai, Y. Shao, C. Hu, S. Lu, S. Ding, High-energy lithium-ion batteries: recent progress and a promising future in applications, *Energy Environ. Mater.* 6 (5) (2023) e12450.
- [5] F.N.U. Khan, M.G. Rasul, A. Sayem, N.K. Mandal, Design and optimization of lithium-ion battery as an efficient energy storage device for electric vehicles: A comprehensive review, *J. Energy Storage* 71 (2023) 108033.
- [6] Z. Zhu, T. Jiang, M. Ali, Y. Meng, Y. Jin, Y. Cui, W. Chen, Rechargeable batteries for grid scale energy storage, *Chem. Rev.* 122 (22) (2022) 16610–16751.
- [7] J. Zeng, S. Liu, Research on aging mechanism and state of health prediction in lithium batteries, *J. Energy Storage* 72 (2023) 108274.
- [8] M.-K. Tran, S. Panchal, T.D. Khang, K. Panchal, R. Fraser, M. Fowler, Concept review of a cloud-based smart battery management system for lithium-ion batteries: Feasibility, logistics, and functionality, *Batteries* 8 (2) (2022) 19.
- [9] M. Kurucan, M. Özbaltan, Z. Yetgin, A. Alkaya, Applications of artificial neural network based battery management systems: A literature review, *Renew. Sustain. Energy Rev.* 192 (2024) 114262.
- [10] S. Singh, P.R. Budarapu, Deep machine learning approaches for battery health monitoring, *Energy* 300 (2024) 131540.
- [11] Y. Che, X. Hu, X. Lin, J. Guo, R. Teodorescu, Health prognostics for lithium-ion batteries: mechanisms, methods, and prospects, *Energy Environ. Sci.* 16 (2) (2023) 338–371.
- [12] S. Doose, A. Hahn, S. Fischer, J. Müller, W. Haselrieder, A. Kwade, Comparison of the consequences of state of charge and state of health on the thermal runaway behavior of lithium ion batteries, *J. Energy Storage* 62 (2023) 106837.
- [13] A. Bavand, S.A. Khajehoddin, M. Ardakani, A. Tabesh, Online estimations of li-ion battery SOC and SOH applicable to partial charge/discharge, *IEEE Trans. Transp. Electrification* 8 (3) (2022) 3673–3685.
- [14] Y. Liu, L. Wang, D. Li, K. Wang, State-of-health estimation of lithium-ion batteries based on electrochemical impedance spectroscopy: a review, *Prot. Control. Mod. Power Syst.* 8 (3) (2023) 1–17.
- [15] J. Ren, J. Ma, H. Wang, T. Yu, K. Wang, A comprehensive review on research methods for lithium-ion battery of state of health estimation and end of life prediction: Methods, properties, and prospects, *Prot. Control. Mod. Power Syst.* (2024).
- [16] P. Li, X. Wu, R. Grosu, J. Hou, M. Iloolov, S. Xiang, Applying neural network to health estimation and lifetime prediction of lithium-ion batteries, *IEEE Trans. Transp. Electrification* (2024).
- [17] J. Lee, J. Won, Enhanced coulomb counting method for SoC and SoH estimation based on coulombic efficiency, *IEEE Access* 11 (2023) 15449–15459.
- [18] L. Chen, S. Xie, A.M. Lopes, H. Li, X. Bao, C. Zhang, P. Li, A new SOH estimation method for lithium-ion batteries based on model-data-fusion, *Energy* 286 (2024) 129597.
- [19] H. Thomas, M.H. Weatherspoon, Capacity and state-of-health prediction of lithium-ion batteries using reduced equivalent circuit models, *Batteries* 11 (4) (2025) 162.
- [20] J. Hou, Y. Jiang, J. Liu, P. Li, S. Xiang, L. Chen, Z. Deng, Adaptive linear time-varying parameter-varying modeling of lithium-ion batteries considering aging phenomenon, *IEEE Trans. Power Electron.* (2025).
- [21] M. Elmahallawy, T. Elfouly, A. Alouani, A.M. Massoud, A comprehensive review of lithium-ion batteries modeling, and state of health and remaining useful lifetime prediction, *IEEE Access* 10 (2022) 119040–119070.
- [22] D. Sun, D. Guo, Y. Lu, J. Chen, Y. Lu, X. Han, X. Feng, L. Lu, H. Wang, M. Ouyang, AI enabled fast charging of lithium-ion batteries of electric vehicles during their life cycle: review, challenges and perspectives, *Energy Environ. Sci.* 17 (20) (2024) 7512–7542.
- [23] M.H. Lipu, S. Ansari, M.S. Miah, S.T. Meraj, K. Hasan, A. Shihavuddin, M. Hannan, K.M. Muttaqi, A. Hussain, Deep learning enabled state of charge, state of health and remaining useful life estimation for smart battery management system: Methods, implementations, issues and prospects, *J. Energy Storage* 55 (2022) 105752.
- [24] S. Ji, J. Zhu, Y. Yang, G. Dos Reis, Z. Zhang, Data-driven battery characterization and prognosis: Recent progress, challenges, and prospects, *Small Methods* 8 (7) (2024) 2301021.
- [25] W. Yao, R. Lai, Y. Tian, X. Li, J. Tian, State of health estimation of lithium-ion batteries using data augmentation and feature mapping, *IEEE Trans. Transp. Electrification* (2024).
- [26] S.-Y. Zhao, K. Ou, X.-X. Gu, Z.-M. Dan, J.-J. Zhang, Y.-X. Wang, A novel transformer-embedded lithium-ion battery model for joint estimation of state-of-charge and state-of-health, *Rare Met.* 43 (11) (2024) 5637–5651.
- [27] D. Wang, C. Li, Z. Wei, X. Chen, W. Wang, A multi-forward-step lithium-ion battery SOH prediction method based on segmented gated recurrent unit and DualMamba, *J. Energy Storage* 117 (2025) 116136.
- [28] H.-K. Wang, M. Gao, X. Dai, L. Cui, A multi-frequency feature extraction and sparse attention mechanism integrated Mamba model for lithium-ion battery state of health estimation, *J. Energy Storage* 123 (2025) 116643.
- [29] R. Zhu, J. Hu, W. Peng, Bayesian calibrated physics-informed neural networks for second-life battery SOH estimation, *Reliab. Eng. Syst. Saf.* 264 (2025) 111432.
- [30] S. Zhang, Z. Liu, Y. Xu, G. Chen, H. Su, An electrochemical aging-informed data-driven approach for health estimation of lithium-ion batteries with parameter inconsistency, *IEEE Trans. Power Electron.* (2025).
- [31] C. Wei, H. Pang, T. Huang, Z. Quan, Z. Qian, Predicting lithium-ion battery health using attention mechanism with Kolmogorov-Arnold and physics-informed neural networks, *Expert Syst. Appl.* (2025) 128969.
- [32] P. Li, Z. Ao, J. Hou, S. Xiang, Z. Wang, Physics-informed mamba neural network with potential knowledge for state-of-charge estimation of lithium-ion batteries, *J. Energy Storage* 123 (2025) 116546.
- [33] Y. Seo, T. Kim, S. Barde, Enhancing battery SOH prediction with Butler-Volmer informed neural networks in data-scarce environments, *Energy* (2025) 138316.
- [34] J. Zhu, Y. Wang, Y. Huang, R. Bhushan Gopaluni, Y. Cao, M. Heere, M.J. Mühlbauer, L. Mereacre, H. Dai, X. Liu, et al., Data-driven capacity estimation of commercial lithium-ion batteries from voltage relaxation, *Nat. Commun.* 13 (1) (2022) 2261.
- [35] F. Wang, Z. Zhai, Z. Zhao, Y. Di, X. Chen, Physics-informed neural network for lithium-ion battery degradation stable modeling and prognosis, *Nat. Commun.* 15 (1) (2024) 4332.
- [36] B. Saha, K. Goebel, Battery data set, 2007, NASA Prognostics Data Repository.
- [37] M. Dubarry, V. Svoboda, R. Hwu, B.Y. Liaw, Incremental capacity analysis and close-to-equilibrium OCV measurements to quantify capacity fade in commercial rechargeable lithium batteries, *Electrochem. Solid-State Lett.* 9 (10) (2006) A454.
- [38] A. Gismero, K. Nørregaard, B. Johnsen, L. Stenhoj, D.-I. Stroe, E. Schaltz, Electric vehicle battery state of health estimation using incremental capacity analysis, *J. Energy Storage* 64 (2023) 107110.
- [39] D. Anseán, V.M. García, M. González, C. Blanco-Viejo, J.C. Viera, Y.F. Pulido, L. Sánchez, Lithium-ion battery degradation indicators via incremental capacity analysis, *IEEE Trans. Ind. Appl.* 55 (3) (2019) 2992–3002.

- [40] C.R. Birkel, M.R. Roberts, E. McTurk, P.G. Bruce, D.A. Howey, Degradation diagnostics for lithium ion cells, *J. Power Sources* 341 (2017) 373–386.
- [41] G. Sun, Y. Liu, X. Liu, A method for estimating lithium-ion battery state of health based on physics-informed machine learning, *J. Power Sources* 627 (2025) 235767.
- [42] F. Wang, Battery-dataset-preprocessing-code-library, 2024, <https://github.com/wang-fujin/Battery-dataset-preprocessing-code-library>.
- [43] Y. Liu, Y. Liu, H. Shen, L. Ding, Battery state of health estimation using a novel BiLSTM-Mamba2 network with differential voltage features and transfer learning, *J. Energy Storage* 110 (2025) 115347.
- [44] H. Sun, D. Yang, L. Wang, K. Wang, A method for estimating the aging state of lithium-ion batteries based on a multi-linear integrated model, *Int. J. Energy Res.* 46 (15) (2022) 24091–24104.
- [45] C.I. Ossai, I.P. Ekwutuoha, Real-time state-of-health monitoring of lithium-ion battery with anomaly detection, Levenberg–Marquardt algorithm, and multiphase exponential regression model, *Neural Comput. Appl.* 33 (4) (2021) 1193–1206.
- [46] N. Yan, X. Li, Y. Zhong, Life decay characteristics identification method of retired power batteries based on inverse power law model of accelerated life test, *Energy Rep.* 8 (2022) 950–956.
- [47] Y. Luo, S. Ju, P. Li, H. Zhang, A method for estimating lithium-ion battery state of health based on physics-informed hybrid neural network, *Electrochim. Acta* 525 (2025) 146110.
- [48] P. Wen, Z.-S. Ye, Y. Li, S. Chen, P. Xie, S. Zhao, Physics-informed neural networks for prognostics and health management of lithium-ion batteries, *IEEE Trans. Intell. Veh.* 9 (1) (2023) 2276–2289.
- [49] L. Chen, C. Chang, X. Liu, J. Jiang, Y. Jiang, A. Tian, Physics-informed neural networks for small sample state of health estimation of lithium-ion batteries, *J. Energy Storage* 122 (2025) 116559.
- [50] A. Kendall, Y. Gal, R. Cipolla, Multi-task learning using uncertainty to weigh losses for scene geometry and semantics, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 7482–7491.
- [51] S. Wang, Y. Teng, P. Perdikaris, Understanding and mitigating gradient flow pathologies in physics-informed neural networks, *SIAM J. Sci. Comput.* 43 (5) (2021) A3055–A3081.
- [52] R. Bischof, M.A. Kraus, Multi-objective loss balancing for physics-informed deep learning, *Comput. Methods Appl. Mech. Engrg.* 439 (2025) 117914.
- [53] F. Wang, Z. Zhai, B. Liu, S. Zheng, Z. Zhao, X. Chen, Open access dataset, code library and benchmarking deep learning approaches for state-of-health estimation of lithium-ion batteries, *J. Energy Storage* 77 (2024) 109884.